:- Data Structure & Algorithms :-

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1. Each node of a singly linked list contains → Data and a pointer to next node.

2. The 'next' field on the last node in a singly linked list stores → NULL

3. A structure that contains a pointer to a variable of its own type is called → Self-referential structure

4. A singly linked list is empty when → head == NULL.

5. Which of the following correctly inserts newnode at the begining of a list with head pointer head → newnode -> next = head;
head = newnode;

6. Which statement sequence correctly delets the first node of a non-empty list → temp = head; head = head -> next; free(temp);

7. Which operation is not directly possible in a singly linked list. → Traversing from tail to head

8. In a singly linked list of n nodes, the total number of next pointer fields is → n

9. Given a pointer p to a node that is not the last node, and no access to the head pointer, the node can be deleted by → copying the data of p->next into p, then deleting p->next

10. The correct loop condition to traverse and print all nodes using a pointer p initialised to head is → while (p != NULL)

11. To insert a node at the end of a singly linked list, traversal must continue until reaching the node where → p->next == NULL

12. Floyd's cycle detection algorithm works by using → Two pointers advancing at different rates

13. Iterative reversal of a singly linked list typically requires how many auxillary pointers → Three

14. The condition head != NULL && head->next == NULL indicates that the list has → Exactly one node.

15. Which C statement correctly allocates memory for one node → p = malloc (struct node);

16. The main purpose of using a dummy (header) node is to → Avoid special case handling for insertion and deletion at the head.

17. In a circular singly linked list, the next field of the last node points to → The first node

18. Which of the following is a genuine disadvantage of a singly linked list compared to an array → It doesn't support random access

19. If the head pointer of a singly linked list is accidentally overwritten without saving it, the result is → Loss of access to all nodes, causing a memory leak.

20. To find the middle node of a singly linked list in a single traversal, one may use → A slow pointer and a fast pointer

21. After executing p = head; while (p-> next != NULL) p = p-> next ; on a non-empty list, the pointer p refers to

→ The last node

22. Which of the following statements is true about a singly linked list → Nodes may be scattered anywhere in memory.

23. Write the C structure declaration for a node of a singly linked list that stores an integer.

→
```
struct node{
    int data;
    struct node* next;
};
```

24. List: 10 → 20 → 30 → 40 → NULL

The loop prints:

10 + 20 = 30
20 + 30 = 50
30 + 40 = 70

Output 30 50 70

25.
```
struct node *insertEnd (struct node *head, int x){
    struct node *t = (struct node *) malloc(size of(struct node));
      t -> data = x;
      t -> next = NULL;
     if (head == NULL)
        return t;
    struct node *p = head;
    while (p -> next ! = NULL)
        p = p -> next;
        p -> next = t;
        return head;
    }
```

26. Write an iterative function to reverse a singly linked list and return the new head pointer

```
struct node * reverse (struct node * head){
    struct node * prev = NULL;
    struct node * curr = head;
    struct node * next;
  while (curr ! = NULL){
     next = curr -> next;
      prev = curr;
      curr = next ;
  }
    return prev;
 }
```

27. Explain, with a diagram, the pointer adjustments required to delete a node that lies between two other nodes in a singly linked list.

→ A → B → C → NULL

To delete B, change the pointer of A:

Before: A → B → C → NULL

After:
A ------- → C → NULL
B (deleted)

Pointer adjustment;

A → next = B → next;
free (B);

28. A singly linked list contains 5 → 8 → 12 → 3 → NULL.

i) Insert 7 at the beginning: 7 → 5 → 8 → 12 → 3 → NULL

ii) Delete the node containing 12: 7 → 5 → 8 → 3 → NULL

iii) Insert 9 at the end; 7 → 5 → 8 → 3 → 9 → NULL